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Counterfactual entanglement distribution using quantum dot spins

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Counterfactual quantum communication allows distant users to share a message without any physical particles traveling through the quantum channel. Here, we present a counterfactual scheme to realize multiparty entanglement distribution. We first realize bipartite counterfactual entanglement distribution to create photon–electron entanglement by using interaction-free measurements and quantum dots. Then the atom–cavity systems are used to create many photon–photon entanglement pairs, which can connect photon–electron pairs with the help of an entanglement beam splitter. Consequently, the electron spins in distant optical microcavities are entangled by the coherent exchange of single flying photons. We also discuss the implementation issues to show that the presented scheme can be realized successfully in experiment. The numerical analysis about the efficiency of multiparty entanglement distribution indicates that there is a trade-off between the entanglement fidelity and the number of users.

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1. INTRODUCTION

Entanglement is a valuable resource for many quantum processing tasks, such as quantum communication [1–4] and computation [5–7]. Therefore, the remote establishment of entanglement, i.e., entanglement distribution, remains an important challenge for practical applications. In general, entangled states should be created from an entanglement source, and then the entangled states are distributed to distant parties. Nevertheless, Zukowski showed that one can realize an "event-ready" Bell–Einstein–Podolsky–Rosen experiment by using independent photons [8]. Subsequently, bipartite [9] and tripartite [10] entanglement swapping were proposed for long-distance entanglement distribution [11–13]. Moreover, Cubitt *et al.* showed that two distant particles can be entangled by sending a third particle that is never entangled with the other two [14]. That is to say, no entangled state is necessary for entanglement distribution. Recently, three research groups have independently implemented entanglement distribution by sending a separable carrier in experiment [15–17]. In addition to that, multiparty entanglement generation has also been researched [18–20]. By using entanglement swapping, multiparty entanglement distribution is allowed among particles belonging to distant users in a communication network.

Counterfactual quantum communication has been proposed recently as a striking phenomenon resulting from quantum mechanics [21]. It can accomplish the communication task without sending any particles through the quantum channel. Counterfactual quantum communication originates from the interaction-free measurement (IFM) proposed by Elitzur and Vaidman [22]. Hosten *et al.* [23] demonstrated counterfactual quantum computation and used a novel "chained" quantum Zeno effect (CQZE) [24] to boost the counterfactual interference probability to unity. Subsequently, Noh [25] proposed a counterfactual quantum cryptography scheme to distribute a secret key with practical security advantages [26,27]. Salih *et al.* [28] presented a direct counterfactual quantum communication protocol to transmit a classical bit directly by using the CQZE system. Guo *et al.* [29] demonstrated a counterfactual scheme to transport an unknown qubit in a nondeterministic manner without prior entanglement sharing or classical communication between two distant participants. Recently, Guo *et al.* [30] proposed a counterfactual protocol for entanglement distribution without any physical particles traveling between two remote users. Then, tripartite counterfactual entanglement distribution was proposed to extend the application scope [31]. Moreover, a counterfactual method to distribute multiparty entanglement was proposed by Shenoy and Srikanth [32]. Unfortunately, the counterfactuality of CQZE was criticized by Vaidman [33–35], who argued that the actual measurement of the presence of the absorption object in the transmission channel contradicts the claims of Hosten [23], Salih [28], and Guo [30]. Furthermore, the application scopes of these counterfactual schemes are limited as they can only be realized by two or three users.

In this work, we present a counterfactual scheme for multiparty entanglement distribution by using an atom–cavity system. We first use interaction-free measurement [36] to realize bipartite counterfactual quantum communication. A charged quantum dot (QD) embedded in an optical resonant cavity is arranged to serve as the absorption object [30]. According to Pauli spin blockade, QD controls the absence and presence of the absorption object by a quantum method. Unlike Guo's scheme [30], the incident photon in our scheme is prepared in a superposition state. Thus, the results in two outputs of IFM are entangled with the absorption object. Due to the partial counterfactuality of IFM [28,34], the protocol aborts if the transmission is not blocked. On the other hand, the protocol is absolutely counterfactual if the transmission is blocked. Consequently, the result of bipartite counterfactual entanglement distribution is a photon–electron entanglement pair. In order to realize multiparty counterfactual entanglement distribution, many photon–photon entanglement pairs are created by atom–cavity systems. Then, the entanglement beam splitter [37] is used to connect the isolated photon–electron entanglement pairs with the help of photon–photon entanglement. In addition, we also discuss the implementation issues and analyze the efficiency of the presented scheme. The numerical analysis results show that the scheme can be realized successfully in experiment.

2. OUR NEW SCHEMES

Here, we first present a counterfactual scheme to realize photon–electron entanglement distribution. Then, by using an atom–cavity system, a photon–photon entanglement pair is created, which can connect two photon–electron entanglement pairs with the help of an entanglement beam splitter.

A. Bipartite Counterfactual Entanglement Distribution

We present the specific procedure of the bipartite counterfactual scheme for distributing photon–electron entanglement. Unlike the existing works, the incident photon and the absorption object used in our scheme are both in superposition states. By using a self-assembled GaAs/InAs quantum dot embedded in an optical microcavity, the blocking or not blocking of the interference is controlled by the quantum absorption object. In the last step, the photon and the absorption object are entangled based on interaction-free measurement. As shown in Fig. 1, our scheme runs as follows:

1. Alice triggers the single-photon source and prepares a single photon with polarization $\frac{|H\rangle+|V\rangle}{\sqrt{2}}$. Then the single photon is sent to the IFM system.

2. The IFM is composed of M beam splitters with large reflectivity $R=\cos^2\theta=\cos^2(\pi/2M)$. The state transformation at the beam splitters is described as $|A\rangle|0\rangle\rightarrow\cos\theta|A\rangle_1|0\rangle_2+\sin\theta|0\rangle_1|A\rangle_2$ and $|0\rangle|A\rangle\rightarrow\cos\theta|0\rangle_1|A\rangle_2-\sin\theta|A\rangle_1|0\rangle_2$, where $|A\rangle=|H\rangle$ or $|V\rangle$, and the subscripts 1 and 2 represent different outputs.

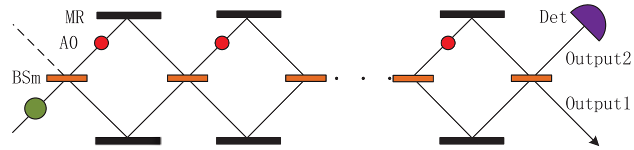

Fig. 1. Principle of bipartite counterfactual entanglement distribution. BS_m, beam splitter with large reflectivity $R=\cos^2\theta=\cos^2(\pi/2M)$; MR, mirror; AO, absorption object, Det, single-photon detector. The state of the incident photon is $\frac{|H\rangle+|V\rangle}{\sqrt{2}}$, and the state of the absorption object is $\frac{|\uparrow\rangle_B+|\downarrow\rangle_B}{\sqrt{2}}$. If the absorption object does not block the transmission, the photon will pass through *Output2* and the detector will click. Otherwise, the photon will pass through *Output1*.

3. According to interaction-free measurement, when Bob allows Alice's photon to pass, the initial state $|A\rangle|0\rangle$ evolves coherently. After m cycles, the photon state can be written as [28]

$$|A\rangle|0\rangle\rightarrow\cos m\theta|A\rangle_1|0\rangle_2+\sin m\theta|0\rangle_1|A\rangle_2. \tag{1}$$

Thus, if $m=M$, the final state will become $|0\rangle_1|A\rangle_2$. On the other hand, when Bob blocks Alice's photon, the final photon state can be written as

$$|A\rangle|0\rangle\rightarrow\cos^{m-1}\theta(\cos\theta|A\rangle_1|0\rangle_2+\sin\theta|0\rangle_1|A\rangle_2). \tag{2}$$

Similarly, if m is large enough, the final state will become $|A\rangle_1|0\rangle_2$.

4. Bob prepares the electron in the quantum dot in the state $\frac{|\uparrow\rangle_B+|\downarrow\rangle_B}{\sqrt{2}}$ to serve as the absorption object. According to Pauli spin blockade [38], if Bob's absorption object is in the state $|\uparrow\rangle_B$, the photon with polarization $|H\rangle$ will be blocked, while the photon with polarization $|V\rangle$ will be transmitted. After m cycles, the photon state can be written as ($|\psi\rangle_A=\frac{|H\rangle+|V\rangle}{\sqrt{2}}$)

$$\begin{aligned}|\psi\rangle_A|\uparrow\rangle_B\rightarrow\frac{1}{\sqrt{2}}&[\cos^{m-1}\theta(\cos\theta|H\rangle_1|0\rangle_2+\sin\theta|0\rangle_1|H\rangle_2)\\&+(\cos m\theta|V\rangle_1|0\rangle_2+\sin m\theta|0\rangle_1|V\rangle_2)]|\uparrow\rangle_B.\end{aligned} \tag{3}$$

Thus, at the end of M cycles, the final state is approximately expressed as $\frac{|H\rangle_1|0\rangle_2+|0\rangle_1|V\rangle_2}{\sqrt{2}}|\uparrow\rangle_B$. Otherwise, if Bob's absorption object is prepared in the state $|\downarrow\rangle_B$, the photon with polarization $|H\rangle$ will be transmitted, while the photon with polarization $|V\rangle$ will be blocked. So the photon state after m cycles is

$$\begin{aligned}|\psi\rangle_A|\downarrow\rangle_B\rightarrow\frac{1}{\sqrt{2}}&[\cos^{m-1}\theta(\cos\theta|V\rangle_1|0\rangle_2+\sin\theta|0\rangle_1|V\rangle_2)\\&+(\cos m\theta|H\rangle_1|0\rangle_2+\sin m\theta|0\rangle_1|H\rangle_2)]|\downarrow\rangle_B,\end{aligned} \tag{4}$$

and the final state is approximately expressed as $\frac{|0\rangle_1|H\rangle_2+|V\rangle_1|0\rangle_2}{\sqrt{2}}|\downarrow\rangle_B$. According to the above analysis, the final state of the system composed of Alice's photon and Bob's absorption object can be written as ($|\phi\rangle_B=\frac{|\uparrow\rangle+|\downarrow\rangle}{\sqrt{2}}$)

$$|\psi\rangle_A|\phi\rangle_B \to \frac{\frac{|H\rangle_1|0\rangle_2+|0\rangle_1|V\rangle_2}{\sqrt{2}}|\uparrow\rangle_B + \frac{|0\rangle_1|H\rangle_2+|V\rangle_1|0\rangle_2}{\sqrt{2}}|\downarrow\rangle_B}{\sqrt{2}}$$
$$= \frac{1}{2}(|H\rangle_1|0\rangle_2|\uparrow\rangle_B + |0\rangle_1|V\rangle_2|\uparrow\rangle_B + |0\rangle_1|H\rangle_2|\downarrow\rangle_B + |V\rangle_1|0\rangle_2|\downarrow\rangle_B). \quad (5)$$

5. As discussed by Salih *et al.* [28], the experimental setup shown in Fig. 1 is only partially counterfactual. In the case when the transmission is not blocked, Eq. (1) implies that the photon passes through the transmission channel with unitary probability. To take the counterfactual interference probability into consideration [36], if given a click at the detector, the probability for finding the photon by a nondemolition measurement of the projection operator on the transmission channel is unity. In other words, we know that the communication is not counterfactual if the photon passes through *Output2*. Therefore, in our scheme, the protocol aborts if the detector clicks. Otherwise, when the photon passes through *Output1*, the system state will collapse to

$$\frac{|H\rangle_1|0\rangle_2|\uparrow\rangle_B + |V\rangle_1|0\rangle_2|\downarrow\rangle_B}{\sqrt{2}} = \frac{|H\rangle_A|\uparrow\rangle_B + |V\rangle_A|\downarrow\rangle_B}{\sqrt{2}}. \quad (6)$$

Obviously, Alice's photon is entangled with Bob's absorption object without any physical particles traveling through the quantum channel.

B. Multiparty Counterfactual Entanglement Distribution

In order to implement multiparty counterfactual entanglement distribution, we use atoms in optical cavities to create photon–photon entanglement pairs, which are very suitable for connecting bipartite counterfactual entanglement distribution. Here, the atom–cavity system allows for a single-photon emission such that the success probability is increased. Another advantage is that the photon state can be stored in an atom for a long time. That offers a clear perspective for scalability and paves the way toward large-scale entanglement distribution.

Assume that multiple users connect with each other in a line, where Alice and Bob are arbitrary adjacent users, as shown in Fig. 2. Each user prepares a charged QD embedded in an optical resonant double-sided microcavity to serve as the quantum absorption object. Then our scheme runs as follows:

1. Alice prepares an atom in the state $|F=2, m_F=0\rangle$. Applying a π-polarized control laser pulse triggers the emission of a single photon and creates the maximally entangled state [39]

$$|\Psi\rangle_{A_a\otimes A'_p} = \frac{1}{\sqrt{2}}(|1,-1\rangle\otimes|R\rangle - |1,1\rangle\otimes|L\rangle) \quad (7)$$

between the spin state of the atom and the polarization of the photon.

2. Alice sends photon A'_p to bipartite counterfactual entanglement distribution presented in the above section. Therefore, if the detector *Output1* clicks, the state of A'_p and B_e will collapse to $|L\rangle_1|\uparrow\rangle$ or $|R\rangle_1|\downarrow\rangle$. If the detector *Output2* clicks, the state of A'_p and B_e will collapse to $|R\rangle_2|\uparrow\rangle$ or $|L\rangle_2|\downarrow\rangle$. In order to achieve counterfactuality, the protocol aborts if the photon passes through *Output2*. Otherwise, the photon passes through *Output1*, which means that the photon is not transmitted

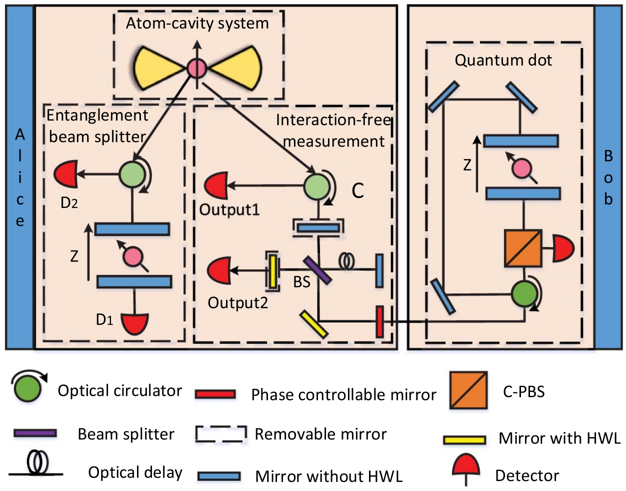

Fig. 2. Experimental setup of multiparty counterfactual entanglement distribution. Each user contains four subsystems: an atom–cavity system, an entanglement beam splitter, an interaction-free measurement, and a quantum absorption object. The quantum absorption object controls whether to block or not block the transmission by a quantum method, while interaction-free measurement is used to realize counterfactual quantum communication. The entanglement beam splitter can create photon–electron entanglement. The atom–cavity system is used to prepare photon–photon entanglement pairs, which can connect photon–electron entanglement pairs. Thus, the electrons in distant quantum dots are entangled without any physical particles traveling through the quantum channel.

through the quantum channel. So the entanglement distribution can be performed with another user when, and only when, the photon passes through *Output1*. However, the probability of passing through *Output1* is only $\frac{1}{2}$. Steps 1 and 2 should be repeated until bipartite counterfactual entanglement distribution succeeds. In this case, the state of the system composed of A_a and B_e can be expressed as

$$|\Phi\rangle_{A_a\otimes B_e} = \frac{|1,-1\rangle_{A_a}|\downarrow\rangle_{B_e} - |1,1\rangle_{A_a}|\uparrow\rangle_{B_e}}{\sqrt{2}}. \quad (8)$$

3. Applying a π-polarized control laser pulse to A_a generates a photon in the polarization state $|\sigma\rangle_{A''_p} = \frac{|L\rangle+|R\rangle}{\sqrt{2}}$, where $|1,-1\rangle_{A_a} \to |1,0\rangle_{A_a}$ with the emission of a single photon $|L\rangle_{A''_p}$, while $|1,1\rangle_{A_a} \to |1,0\rangle_{A_a}$ with the emission of a single photon $|R\rangle_{A''_p}$ [40,41]. Thus, the entanglement state of A''_p and B_e can be written as

$$|\Theta\rangle_{A''_p\otimes B_e} = \frac{|L\rangle_{A''_p}|\downarrow\rangle_{B_e} - |R\rangle_{A''_p}|\uparrow\rangle_{B_e}}{\sqrt{2}}. \quad (9)$$

4. The entanglement beam splitter [37] described in Fig. 2 can be used to entangle A''_p and A_{electron}. Similar to the discussion on bipartite entanglement distribution, the final system state can be expressed as ($|\sigma\rangle = \frac{|L\rangle+|R\rangle}{\sqrt{2}}$, $|\omega\rangle = \frac{|\uparrow\rangle+|\downarrow\rangle}{\sqrt{2}}$)

$$|\sigma\rangle_{A''_p}|\omega\rangle_{A_e} \to \frac{\frac{|L\rangle_1+|R\rangle_2}{\sqrt{2}}|\uparrow\rangle + \frac{|L\rangle_2+|R\rangle_1}{\sqrt{2}}|\downarrow\rangle}{\sqrt{2}}$$
$$= \frac{|L\rangle_1|\uparrow\rangle + |R\rangle_2|\uparrow\rangle + |L\rangle_2|\downarrow\rangle + |R\rangle_1|\downarrow\rangle}{2}, \quad (10)$$

where the subscripts 1 and 2 represent different paths. Thus, if the detector D_1 clicks, the entangled state of A_p'' and A_e will collapse to $|L\rangle_1|\uparrow\rangle + |R\rangle_1|\downarrow\rangle$. In this case, the distributed entanglement will be $|\uparrow\rangle_{A_e}|\downarrow\rangle_{B_e} - |\downarrow\rangle_{A_e}|\uparrow\rangle_{B_e}$. However, if the detector D_2 clicks, the entangled state of A_p'' and A_e will collapse to $|R\rangle_2|\uparrow\rangle + |L\rangle_2|\downarrow\rangle$, and the distributed entanglement will be $|\uparrow\rangle_{A_e}|\uparrow\rangle_{B_e} - |\downarrow\rangle_{A_e}|\downarrow\rangle_{B_e}$. Alice broadcasts the measurement results to other users such that the distributed entanglement state is shared.

5. Assume that Bob's other neighbor is Charlie. The next step is to entangle Bob's electron B_e and Charlie's electron C_e. In the same way, Bob repeats steps 1 and 2 until $|\Phi\rangle_{B_a \otimes C_e}$ is created successfully (photon B_p' passes through *Output1*). Thus, the success probability of bipartite entanglement distribution does not affect the efficiency of multiparty entanglement distribution. Then steps 3–5 are performed to entangle B_e and C_e. Consequently, A_e, B_e, and C_e are all entangled without any physical particles traveling through the quantum channel. Repeat the above steps until all the electron spins in distant quantum dots are entangled.

3. DISCUSSION

Presently, the term "counterfactual" is still a subject of heated debate. In accordance with Vaidman's opinions, CQZE is only partially counterfactual [33,34]. In the case when there is no blockade, the possibility for finding the photon by a nondemolition measurement on the transmission channel is unity. Then the scheme proposed by Guo *et al.* [30] fails to distribute entanglement with unitary counterfactual interference probability. Here, we use an incident photon with superposition state and IFM to realize counterfactual entanglement distribution. Due to the principle of IFM, the result expressed as Eq. (5) is partially counterfactual. When the photon passes through *Output2*, we know that the transmission is not blocked. So the protocol aborts if the corresponding detector clicks. Thus, Eq. (6) is the result when the transmission is blocked. Moreover, the counterfactuality of Eq. (6) has also been approved by Vaidman. That is to say, our scheme is absolutely counterfactual no matter what the debate result of the counterfactuality of CQZE is.

Recently, Guo *et al.* proposed a scheme for a distributed controlled-phase gate between two distant quantum-dot electron-spin qubits in optical microcavities [30]. Their scheme can also be used to implement entanglement distribution between two remote users. However, their scheme still uses CQZE to boost the counterfactual interference probability to unity. As discussed above, the counterfactuality is questioned [33]. Furthermore, the photon loss rate in IFM is a non-negligible influence factor [30]. When Guo's scheme is extended to multiparty, the photon loss between two nodes would affect the whole entanglement fidelity. However, our scheme uses an atom–cavity system to separate the IFM and the entanglement beam splitter. As discussed in multiparty counterfactual entanglement distribution, Steps 1 and 2 can be repeated until bipartite counterfactual entanglement distribution succeeds. Obviously, the success probability of our scheme is higher than that of Guo's scheme. Thanks to the separation of the IFM system and the entanglement beam splitter, bipartite counterfactual entanglement distribution between two different adjacent users

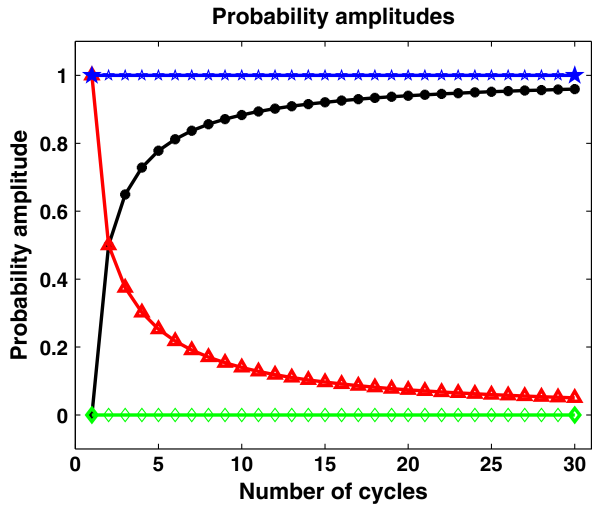

Fig. 3. Probability amplitudes of each superposition in Eq. (11) versus the value of cycles. The blue line represents the probability amplitude of $|0\rangle_1|A\rangle_2$ and the green line represents the probability amplitude of $|A\rangle_1|0\rangle_2$ when the photon is not blocked, while the red line represents the probability amplitude of $|0\rangle_1|A\rangle_2$ and the black line represents the probability amplitude of $|A\rangle_1|0\rangle_2$ when the photon is blocked.

can be performed simultaneously. After that, the entanglement beam splitter is used to connect bipartite entanglement distribution. Taking the execution time into consideration, our scheme is also efficient. Another advantage is that the atom–cavity system allows for a single-photon emission, which accordingly increases the success probability [39].

Now, we begin to discuss the performance of the presented counterfactual scheme. The entanglement distributed by interaction-free measurement is approximately maximally entangled. As the quantum absorption object is prepared in superposition state, the incident photon with polarization $|H\rangle$ or $|V\rangle$ would be transmitted and reflected with equal probability. According to Eqs. (1) and (2), the final entangled state shared by two users will become

$$|\Phi\rangle_m \to \frac{1}{\sqrt{2}}[\cos m\theta|A\rangle_1|0\rangle_2 + \sin m\theta|0\rangle_1|A\rangle_2 + \cos^{m-1}\theta(\cos\theta|A\rangle_1|0\rangle_2 + \sin\theta|0\rangle_1|A\rangle_2)], \tag{11}$$

where $m = M$ and $\theta = \pi/2M$. As shown in Fig. 3, we plot the variation trend of the coefficients with the values of m (number of cycles). It is obvious that $\cos m\theta = 0$ and $\sin m\theta = 1$ for any values of m. For large values of m, $\cos^m\theta \to 1$ and $\cos^{m-1}\theta \sin\theta \to 0$, for example, for $m = 50$, $\cos^m\theta = 0.9756$ and $\cos^{m-1}\theta \sin\theta = 0.0307$. After the interaction-free measurement, the state in Eq. (11) can be approximately expressed as $|\Phi\rangle_m \simeq \frac{1}{\sqrt{2}}(|A\rangle_1|0\rangle_2 + |0\rangle_1|A\rangle_2)$.

The perfect entanglement requires that the state of the photon–electron system becomes a maximally entangled state, i.e., $|\Phi'\rangle_m = \frac{1}{\sqrt{2}}(|A\rangle_1|0\rangle_2 + |0\rangle_1|A\rangle_2)$. Moreover, our scheme only uses the results when the photon passes through *Output1*. Hence, the fidelity of each bipartite counterfactual entanglement

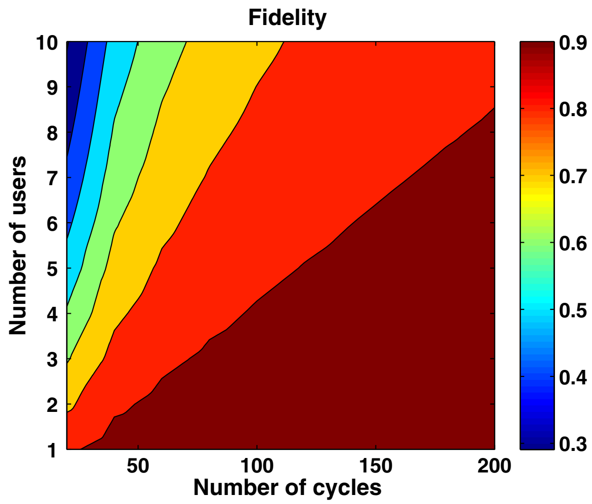

Fig. 4. Fidelity of entanglement versus the value of cycles and the number of users. The number of cycles affects the fidelity of entanglement distributed by elementary links, while the number of users affects the fidelity of final target entanglement shared by multiparty entanglement. The fidelity of distributed multiparty entanglement would decrease when the number of cycles decreases or the number of users increases.

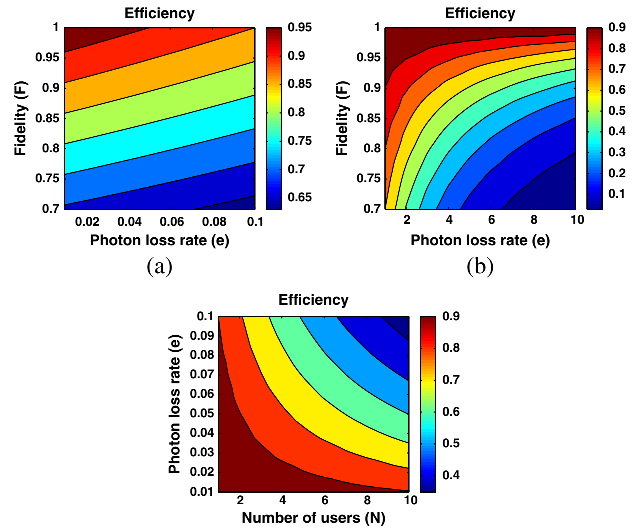

Fig. 5. Efficiency of entanglement distribution versus different values of photon loss rate e, entanglement fidelity F, and number of users N. (a) The efficiency of entanglement distribution between two adjacent users would increase when the photon loss rate decreases or the fidelity increases. (b) The efficiency of multiparty entanglement distribution would increase when the fidelity increases or the number of users decreases. (c) The efficiency of multiparty entanglement distribution would decrease when either the photon loss rate or the number of users increases.

distributed by IFM can be obtained as $F = |_m\langle\Phi'|\Phi\rangle_m|^2 = (\cos^m\theta)^2$. Similarly, the result of N-party counterfactual entanglement distribution is in the form of $|\Psi\rangle = \frac{1}{\sqrt{2}}[(\cos^m\theta)^N |\alpha_1\rangle|\alpha_2\rangle\ldots|\alpha_N\rangle + (\cos^m\theta)^N|\beta_1\rangle|\beta_2\rangle\ldots|\beta_N\rangle]$, where $|\alpha_i\rangle$ and $|\beta_i\rangle$ are the superposition states of the ith user. For N users, the fidelity of the shared entanglement state is $F_N = \frac{1}{4}[(\cos^m\theta)^N + (\cos^m\theta)^N]^2 = (\cos^m\theta)^{2N}$. As shown in Fig. 4, there is a trade-off between the target fidelity and the number of users.

Due to the photon attenuation in the transmission channel, the photon loss rate will affect the efficiency of the presented scheme. In particular, the scheme requires multiple particles to be entangled so that the photon loss is a non-negligible influence factor. Suppose that the photon loss rate in every IFM is e. Thus, the success probability of our scheme will be $(1 - e)^N$. As discussed above, the photon state will collapse to *Output1* with probability $\frac{1}{2}$, which means that the photon is not transmitted through the quantum channel. However, thanks to the atom–cavity system, the processes of the IFM and the entanglement beam splitter are separated. The IFM is first required to be performed repeatedly until it succeeds. Then the entanglement beam splitter is applied to connect the entanglement pairs. Therefore, single IFM does not affect the success probability of multiparty entanglement distribution, but it affects the execution time. For simplicity, we do not take the efficiency of IFM into consideration. Thus the efficiency of the presented scheme is $E_M = (1 - e)^N F^N = [(1 - e)F]^N$. As shown in Fig. 5, the success probability is reduced as either the number of users or photon loss rate increase. Fortunately, the photon loss rate affects the efficiency, not the fidelity, of distributed entanglement.

In a quantum dot, the electron-spin decoherence would decrease the fidelity. In an atom–cavity system, the atom spin is also required to be shielded from the environment long enough, i.e., having long coherence time. Suppose that the average distance between two nodes is L. Then the required time to accomplish the scheme will be $t \approx mL/c$ (c is the speed of light). Thus, the spin coherence time T_c should satisfy $T_c \geq t$. If the presented scheme performs N-particle entanglement distribution, the spin coherence time should satisfy $T_c \geq Nt$. Hence, for given T_c and N, the distance of the entanglement distribution is limited. As shown in Fig. 6, there is a trade-off between entanglement fidelity and distribution distance. Prolonging the coherence time is a main challenge

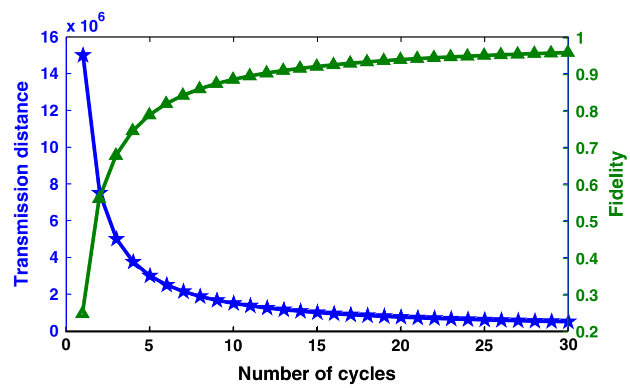

Fig. 6. Fidelity and transmission distance of bipartite entanglement versus the number of cycles. The green line represents the fidelity of bipartite entanglement, which increases when the number of cycles increases. The blue line represents the transmission distance of bipartite entanglement, which decreases when the number of cycles increases.

for realizing the presented multiparty counterfactual entanglement distribution in practical applications. Fortunately, more and more works on suppressing the nuclear spin fluctuations effectively and prolonging the atom spin coherence have been conducted [42–45].

4. METHODS

We begin to discuss the implementation issues of the presented scheme. Figure 2 shows the experimental setup of our counterfactual entanglement distribution scheme between two adjacent users. First, we need to use an atom–cavity system to generate a photon–photon entanglement pair. One photon is sent to IFM for entangling with Bob's absorption object with unitary counterfactual interference probability, while another photon is sent to the entanglement beam splitter, which is used to entangle with Alice's electron in the quantum dot. Thus, Alice's electron and Bob's electron in their quantum dots are entangled successfully.

Here, we can use a single ^{87}Rb atom coupled with an optical cavity, which is prepared in the state $|F = 2, m_F = 0\rangle$ of the $5S_{1/2}$ ground level [39]. A π-polarized laser with the transition from $F = 2$ to $F' = 1$ of the excited $5P_{3/2}$ level, together with the cavity (coupling level $F = 1$ and $F' = 1$), drives a vacuum-stimulated Raman adiabatic passage to the $|F = 1\rangle$ state of the ground level. Two different paths to the state $|1, \pm 1\rangle = |F = 1, m_F = \pm 1\rangle$ are possible, resulting in the generation of a σ^- or σ^+ photon, respectively. After a single-photon emission, the entangled state of the system is shown in Eq. (7). Similarly, applying a π-polarized laser pulse to the atom in the superposition state $\frac{|F=1,m_F=1\rangle+|F=1,m_F=-1\rangle}{\sqrt{2}}$ can generate a single photon in the superposition state $\frac{|L\rangle+|R\rangle}{\sqrt{2}}$. Consequently, a photon–photon entanglement pair is created. As the atom–cavity system can generate a single photon in a well-defined mode very efficiently, such a state mapping is feasible.

Recently, Hu *et al.* proposed a singly charged self-assembled GaAs/InAs quantum dot being embedded in an optical resonant double-sided microcavity [37], which has been recognized by Bonate *et al.* [46]. Constructing a quantum version of the absorption object, as shown in Fig. 2, the incident photon first passes through a polarizing beam splitter in the circular basis (optical circulator), which transmits the right circularly polarized photon and reflects the left circularly polarized photon. If the photon is not blocked, the photon would pass through the cavity. Otherwise, if the photon is blocked, both the polarization and the propagation direction of it will be flipped. Then the photon is reflected by the optical circulator and absorbed by the detector. In summary, the incident photon with polarization $|R\rangle$ will be transmitted when the electron spin is in the state $|\uparrow\rangle$ and reflected by the cavity when the electron is in state $|\downarrow\rangle$. Similarly, a photon with polarization $|L\rangle$ will be transmitted when the electron spin is in the state $|\downarrow\rangle$ and is reflected by the cavity when the electron spin is in the state $|\uparrow\rangle$ [30]. Therefore, the quantum-dot–microcavity system is very suitable for serving as the quantum version of the absorption object.

Moreover, Hu's scheme is also used as an entanglement beam splitter in our scheme. As shown in Fig. 7, the incident photon is prepared in the superposition state $\frac{|R\rangle+|L\rangle}{\sqrt{2}}$, while the electron in the quantum dot is prepared in the superposition state $\frac{|\uparrow\rangle+|\downarrow\rangle}{\sqrt{2}}$. As discussed in bipartite counterfactual entanglement distribution, if the photon is transmitted and passes through path 1, the entangled state is $\frac{|R\rangle|\uparrow\rangle+|L\rangle|\downarrow\rangle}{\sqrt{2}}$, while the entangled state is $\frac{|R\rangle|\downarrow\rangle+|L\rangle|\uparrow\rangle}{\sqrt{2}}$ if the photon is reflected by the cavity and passes through path 2. Therefore, we can know the distributed entanglement results according to the detectors' clicks.

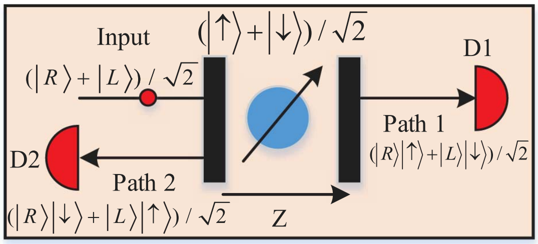

Fig. 7. Experimental setup of entanglement beam splitter. The incident photon is in the state $\frac{|R\rangle+|L\rangle}{\sqrt{2}}$. The electron in the quantum dot is prepared in the state $\frac{|\uparrow\rangle+|\downarrow\rangle}{\sqrt{2}}$. D_1 and D_2 are single-photon detectors.

5. CONCLUSION

In conclusion, we present a multiparty entanglement distribution scheme without any physical particles traveling through the quantum channel. Theoretically, our scheme is more secure than the existing entanglement distribution schemes because of counterfactuality. Furthermore, the application scope is extended as our scheme distributes entanglement among multiple users. The numerical analysis results show that the present scheme is feasible in experiment, which makes commercial applications conceivable.

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